

Will You Ride Google's Wave?

How would a 'unified' communication platform look like? It would look like Google Wave—an e-mail service, an online collaboration tool, or a real-time communication stream, all bundled onto one platform

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Say hello to Google Wave, a new communication tool that has the potential to be the next big thing after e-mail and instant messaging—the two most spectacular successes in digital communication.

The new technology glues together a number of things—including e-mail, instant messaging, blogging, document sharing, wikis and multimedia content—which have been until now separate products. In other words, Wave is part social networking, part unified communication—and all Google.

A wave is shared

"In Google Wave, you can create a Wave and add people to it. Everyone on your Wave can use formatted text, photos, gadgets and even feeds from other sources on the Internet. They can insert a reply or edit the wave directly. It is concurrent rich-text editing, where you can see instantly what your fellow collaborators are typing in your Wave. That means Google Wave is just as well suited for quick messages as for persistent content—it allows for both collaboration and communication. "You can also use 'playback' to rewind the Wave and see how it evolved," says Lars Rasmussen, software engineering manager, Google.

Let's put it in simple terms. For instance, you are going to Switzerland on a holiday and want to know about the most recommended places in the country. Now all you have to do is create a new wave and ask friends for advice on places to visit. They could add their suggestions to the Wave, or even invite other friends to share their views. Users could also share pictures, videos and maps across the Wave. (You can

Now communicate and collaborate in real time

With Wave, you can do live editing of rich text document, share photos as you do in an online photo gallery, do an instant messaging type of communication and live transmission inside a single application.

also directly drag items from your PC—like photos—into the Wave.)

And, in case you have just joined the Wave and want to pull up the entire conversation, the playback function can help you out. It allows new participants to play the Wave stream back post-by-post.

Wave has been constructed on Google Web Toolkit using HTML 5 and has a rich set of open APIs that allow developers to embed waves in other Internet services and to build extensions that work inside them. This means developers can use and enhance Google Wave through two types of developments: By building robot extensions to automate common tasks or building gadget extensions to provide a new way for users to interact.

A wave is live

The Google Wave protocol is the underlying format for storing and sharing of waves, and includes the 'live' concurrency control, which allows edits to be reflected instantly across users and services.

Although Google Wave works like an e-mail, it does not send a message along with its entire thread of previous messages, or require all responses to be stored in each user's inbox for context. Instead message documents, referred to as waves, contain complete threads of multimedia messages (blips) and get

perpetually stored on a central server. Waves are shared with collaborators who can be added to or removed from the wave at any point during a wave's existence.

Google Wave is still in active development and is expected to remain in development until early 2010.

Open sourcing Wave

As with Android, Google Chrome and many other Google efforts, the company plans to make the code open source to encourage the developer community to get involved.

Google has already open sourced some components of Wave, namely the operational transform (OT) code, the underlying wave model and a basic client/server prototype that uses the wave protocol. The company is promoting the developers' interest for implementations and hence making it easier for users and businesses to customise applications.

"The OT code is the heart and soul of the collaborative experience in Google Wave, and we plan that code will evolve into the production-quality reference implementation," says Google.

The prototype is intended primarily as a "simple 'hello, world' implementation" to encourage experimentation using the Google Wave Federation Protocol. All of this code—nearly 40000 lines of Java code—is available under the Apache 2.0 licence. Google has also announced that the protocol specification, white papers and API documentation are being made available under a Creative Commons Attribution licence.

Google plans to open source more code as Wave evolves. ●

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